

CHARCUBE · REQUIREMENTS

# CharCube requirements

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DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
| Document ID | CCB-REQ-001                           | Version | 0.2            |
| Status      | Draft                                 | Date    | 2026-09-25     |
| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 2 Technology concept formulated   |         |                |
| Repository  | github.com/BoujeeEnjinia1701/charcube |         |                |
| Source      | docs/03-requirements.md @ 7440748     |         |                |

REVISION HISTORY

| Version | Date       | Author       | Change                                                                             |
|---------|------------|--------------|------------------------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                   |

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These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be checked by calculation at TRL 3 and revised after co-design sessions. The status column compares each target with the estimates in CCB-PRC-001; "unverified" means the concept cannot yet show it either way.

| ID  | Requirement                          | Target                                                                                                                                                                                                                                                         | Verification (TRL 3 or later)                                | Status at TRL 2                                                                                                 |
|-----|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| R1  | Process a useful batch               | 10 kg or more of air-dry residue (15 % moisture or less) per batch                                                                                                                                                                                             | Retort volume and packed bulk density                        | Met on estimate for bundled straw and stalks (about 12 kg); <b>not met</b> for loose chopped straw (about 4 kg) |
| R2  | Make biochar efficiently             | Biochar 25 % or more of dry feed mass                                                                                                                                                                                                                          | Mass balance; later weighed batches                          | At risk: estimate 28 %, range 20 to 35 %                                                                        |
| R3  | Make stable char                     | Retort core at 450 °C or more for 30 min or more per batch, logged; target molar H/Corg below 0.7, the <a href="#">European Biochar Certificate</a> limit                                                                                                      | Heat balance; later logged batches and lab analysis          | Unverified                                                                                                      |
| R4  | Burn the gas, not vent it            | All retort gas passes the annulus fire and the burner throat; visible smoke only during the first 15 min after lighting and the last 10 min of a batch; methane below 24 g per kg of char (the retort field average in <a href="#">Sparrevik et al. 2015</a> ) | Design review; later emission measurement with a partner lab | <b>Not shown:</b> routing is met by design, emissions cannot be shown at TRL 2                                  |
| R5  | Fit a daily routine                  | Light to end of flaming in 4 h or less; char ready to unload within 16 h of lighting                                                                                                                                                                           | Heat balance; later logged batches                           | Met on estimate (2.5 to 3.5 h burning, 10 to 14 h cooling)                                                      |
| R6  | Recover heat as hot water            | 10 MJ or more into water per batch (60 L raised by 40 K)                                                                                                                                                                                                       | Heat transfer estimate; later measured water temperature     | Met on estimate (about 15 MJ, range 10 to 25 MJ)                                                                |
| R7  | Keep the water circuit safe          | Jacket open to the air at all times; no valve or cap that can seal it; water drawn from a tap at the base; jacket kept away from the lid by a separate support                                                                                                 | Design review and safety checklist                           | Met by design                                                                                                   |
| R8  | Use little start-up fuel             | 5 kg or less of dry wood per batch                                                                                                                                                                                                                             | Heat balance; later weighed batches                          | At the limit (about 5 kg); unverified                                                                           |
| R9  | Record each batch                    | Two type K channels (retort core, throat exit), 0 to 1,000 °C, $\pm 5$ °C or better, logged every 10 s to removable storage, 24 h on a USB power bank                                                                                                          | Component data; design review                                | Met by design                                                                                                   |
| R10 | Stay within the concept budget       | Parts \$300 or less                                                                                                                                                                                                                                            | Priced BOM                                                   | Met with about \$1 margin (about \$299); safety equipment not included                                          |
| R11 | Be built and handled by a small crew | Built with hand tools, drill and angle grinder, with at most one welded part (the water jacket); no single lift above 25 kg per person                                                                                                                         | Design review; massing model                                 | Met only if the jacket is drained and the 36 kg tripod unit is lifted by two people                             |
| R12 |                                      |                                                                                                                                                                                                                                                                |                                                              |                                                                                                                 |

| ID | Requirement             | Target                                                                                                                               | Verification (TRL 3 or later)      | Status at TRL 2                    |
|----|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------|
|    | Operate safely outdoors | Flue outlet 2.5 m or more above the ground; 5 m clear radius from structures and dry vegetation; no galvanized parts in the hot path | Design review and safety checklist | Met by design (outlet about 2.7 m) |

## ■ Requirements not met or not yet shown

- **R1** is not met for loose chopped straw: at about 40 kg/m<sup>3</sup> the retort holds only about 4 kg. Straw must be bundled or packed, or mixed with stalks and cobs.
- **R4** cannot be shown at TRL 2. The gas route is closed by design, but methane and smoke depend on burner tuning and need measurement.
- **R2, R3 and R8** are estimates with wide ranges and depend on feedstock and operator.
- **R10** leaves no margin and excludes a fire extinguisher, gloves and eye protection (about \$40).
- **R11** depends on two people lifting the drained tripod unit.

## ■ Assumptions

- Feedstock: air-dry cereal straw, maize or cotton stalks, cobs or prunings at 10 to 15 % moisture, packed to about 120 kg/m<sup>3</sup>, higher heating value about 15 MJ/kg.
- Start-up fuel: about 5 kg of dry wood at about 16 MJ/kg, preferably from residues or prunings.
- Char: about 24 MJ/kg and about 55 % organic carbon for mixed straw and stalks (IPCC 2019 defaults are 49 % for rice straw char and 65 % for herbaceous material).
- One batch a day for about 250 days a year.
- Ambient 10 to 40 °C, dry weather, wind below about 5 m/s.
- The kiln serves a household's or a trial plot's residue, not a whole field (see CCB-PRB-001, out of scope).